Seeing what you eat

Can a nutrition app tell when it is guessing? Final results.

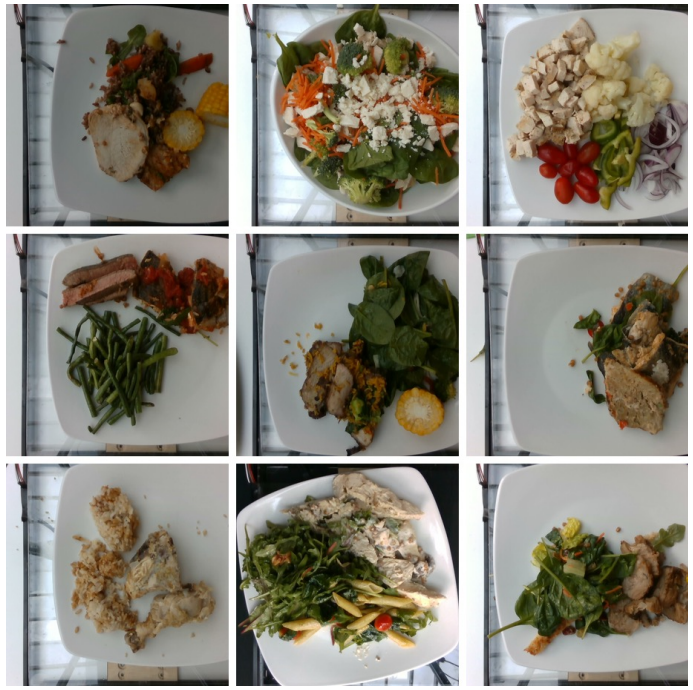
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COMPSCI 760 Advanced Machine Learning · Group 1 · 19 October 2026

Code and results **github.com/sushantkrishnan/seeing-what-you-eat**

MOTIVATION

Every calorie app answers every photo.



Nutrition5k dishes [1]: cafeteria meals, every ingredient weighed.

71 kcal is our model's average miss, on meals that average 255 kcal. The typical miss is smaller; the trouble is the tail. [ours]

253 kcal is the average miss on its worst tenth of photos: about a whole meal. That tenth carries 36% of all the error, and the app shows those numbers with the same confidence as the rest. [ours]

29% less error if the app could decline exactly those photos and answer the other nine in ten (70.9 → 50.4 kcal). [ours]

An app that knew when it was guessing could ask "please check this one" instead. The latest nutrition benchmark asks for exactly this: "assessing reliability under degraded conditions" [2].

PROBLEM AND OBJECTIVES

We froze the predictor and studied the gate.

PROBLEM

Before it shows a number, an app has to decide: answer, or ask for help. It needs a signal that ranks photos by how wrong the answer will be, without knowing the truth.

OUR APPROACH

Keep the predictor fixed (it matches the published baseline, slide 6) and benchmark the gate: every signal an app could use, from free pixel checks to model-aware scores, on the error-versus-reject protocol [3], between refusing at random and a perfect refuser.

OBJECTIVES, AND WHERE THEY ENDED

O1 **Benchmark gate signals on Nutrition5k, three backbones.**

Done: ten signals, three more tried since the update. After our first results the main test moved from damaged photos to clean ones: the dish drives the error.

O2 **Show which wins are real, with intervals.**

Done. Session-level intervals and a size control; since the update, four checks of whether the result holds.

O3 **Honest ranges on the photos the app answers.**

Done. Size-aware ranges, checked the same way.

BORROWED VS OURS Borrowed: the data and split [1], frozen backbones, error-versus-reject [3], the corruption ladder [4], the gate signals [6]–[9], conformal ranges [5], [11], [12]. Ours: the benchmark and its size control, plate sessions as the unit, the confirmation checks, and every result in this talk.

RESPONSE TO FEEDBACK

You asked if it was conclusive. We checked.

AFTER THE PROPOSAL

Asked which evaluation metrics we would use and how we would show significance.

One fixed protocol: error at 90% answered and over the whole curve, 95% intervals from a bootstrap over plate sessions, paired tests against random with a Holm correction, and a control that reads only the predicted meal size.

AFTER THE UPDATE

“Are the results inconclusive? With p = 0.05 it could go either way if we had more data.”

We had claimed that inside size bands the error head cuts the error by 6.5 kcal ($p < 0.01$). We tested that claim four ways, tried two exploratory ideas, and finished the ranges.

FOUR CHECKS: DOES THE CLAIM HOLD?

1 **New data** 367 plate sessions never used to score a gate; the hypotheses were written down first.

2 **Retrain 20 ways** change which photos calibrate the model, and watch the result.

3 **One pooled estimate** all 3,262 dishes, each scored by models that never saw it, 5 refits averaged.

4 **Placebo** 200 error heads trained on shuffled errors: what does luck look like?

ALSO NEW SINCE THE UPDATE

1 **A steadier error head** average 10 heads, each calibrated on different photos. Exploratory.

2 **Disagreement across backbones** do three different models disagree on hard photos? Exploratory.

3 **Size-aware ranges** objective 3: ranges that widen with the meal, checked the same way as the gates.

DATASET AND PREPROCESSING

Nutrition5k: weighed meals, scanned in sessions.

4,768
dishes in the official split [1]

→

3,490
have an overhead photo

→

2,755 / 507
train / test, official split

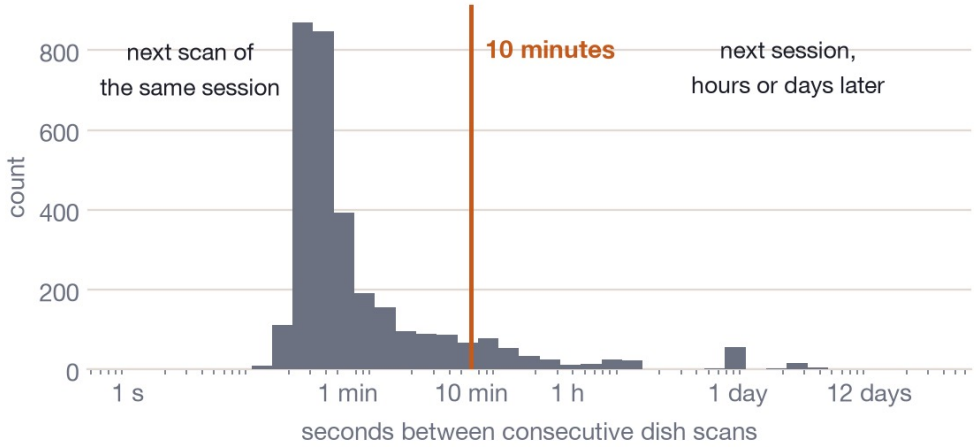
→

1,924 / 831
fit / calibration, carved by session

→

367 / 127
plate sessions, train / test

SCANS OF ONE PLATE ARRIVE ~40 S APART



Errors within a session are correlated (intraclass correlation 0.34), so the session is our unit for every split and interval. Train and test scans interleave in time; pooled, they chain into 366 clusters, which we use as the unit there.

PROBLEMS WE HIT, AND WHAT WE DID

Ragged metadata. Variable-length rows; we parse them by anchoring on ingredient ids.

Missing photos. Only 3,490 dishes have an overhead photo, so 2,755 / 507 are usable.

Labels are exact sums of weighed ingredients. That makes a perfect refuser computable. 83 test dishes are under 50 kcal, so percentage error is unstable: we score absolute kcal.

No real bad photos. Eight simulated damage levels [4]; our “phone” level is an imitation, and labelled as one.

Why this dataset. It is the largest public set we found with weighed, per-dish calories [1].

METHOD

One frozen model, many signals, one decision.



THE SIGNALS A GATE CAN USE

No model	photo sharpness (Laplacian variance [16]), Tenengrad, JPEG blockiness, high-frequency ratio
Distance	from the training photos: Mahalanobis [6], nearest neighbours [7]
Disagreement	spread of 32 bootstrap heads [8]
Learned	an error head that predicts |error| from the features [9]
Control	the predicted calorie count alone (ours): does a gate just refuse big meals?

BASELINE: CALORIES, 507 TEST DISHES

model	MAE	source
Inception V2, trained	**70.6**	[1]
Ours: CLIP, frozen	**70.9**	[ours, replicated]
Ours: SigLIP 2 / DINOv3	**67.5 / 63.2**	[ours]
Best published, needs depth	**41.3**	[1]

Our CLIP interval 62.0–80.4 contains the published 70.6, with no training. MAE: mean absolute error, kcal.

A gate only has to rank photos by how wrong they will be. The weighed truth tells us every photo's real error, so we can score that exactly.

EXPERIMENTAL DESIGN

Every split has one job.

split	size	its only job
fit	1,924 dishes	fit the ridge heads and the gate signals
calibration (validation)	831 dishes	pick the ridge penalty; train the error head; set the size bands and range width
test	507 dishes, 127 sessions	scored, never tuned on
new sessions (check 1)	2,755 dishes, 367 sessions	out-of-fold rerun on train, hypotheses written first
pooled (check 3)	3,262 dishes, 366 clusters	5-fold cross-fit, 5 refits averaged

MEASURES

MAE on the photos the app answers, at 90% answered and as the area under the whole risk–coverage curve (AURC [10]), against refusing at random and a perfect refuser; again inside 10 predicted-size bands. Ranges: coverage by meal size and the interval score [18].

STATISTICS

95% intervals from 2,000 bootstrap resamples of plate sessions, paired across gates; Holm correction across the gates [17]; placebo heads trained on shuffled errors.

TUNING

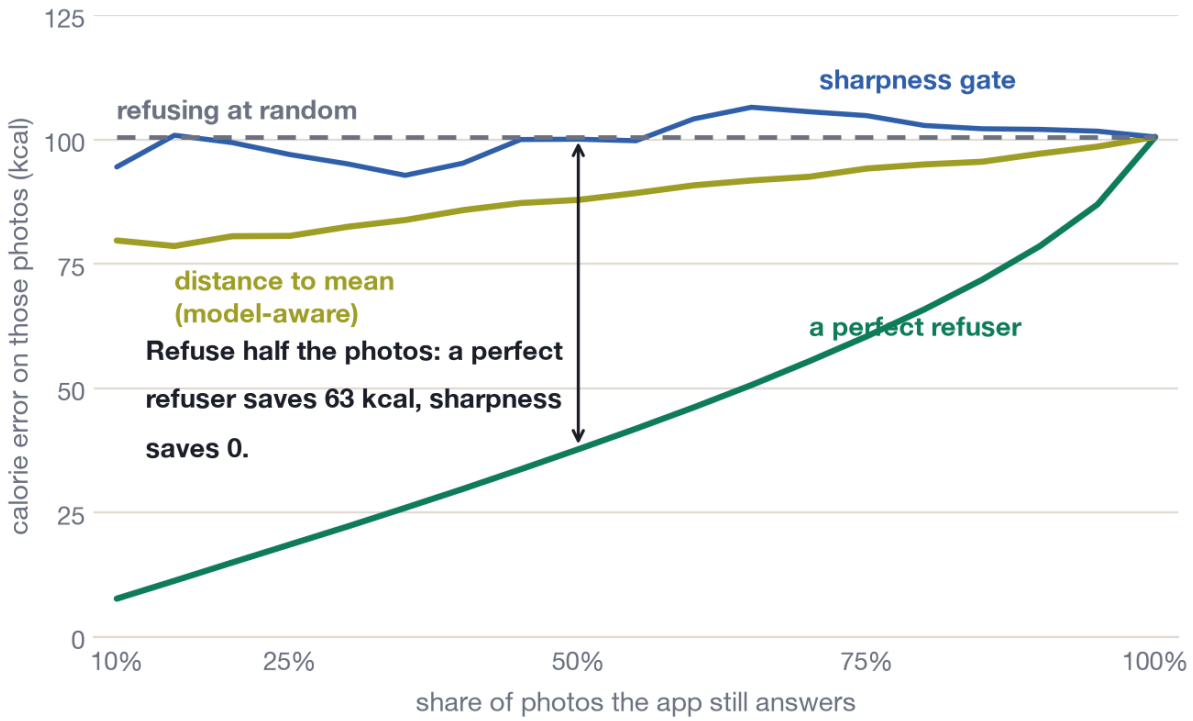
One knob, the ridge penalty, picked on calibration from a 25-value grid. Fixed in advance: 64 principal components, 10 neighbours, 32 heads; 10 size bands (deciles; 5 and 20 also run); 90% answered (an app should rarely refuse; the whole curve is reported too).

COMPUTE

One M-series laptop. Features at ~68 photos/s, computed once. Every head is closed form, so there is no training loop to converge. The whole study reruns in ~100 min: reproduce.sh.

RESULT 1

Photo quality is not the warning sign.



507 test dishes × 8 damage levels, CLIP [ours]. Error against the share of photos answered; olive, for contrast: a model-aware signal.

8.5%

of the error variance comes from photo damage; 62% comes from which dish it is. [ours]

AUC ≈ 1

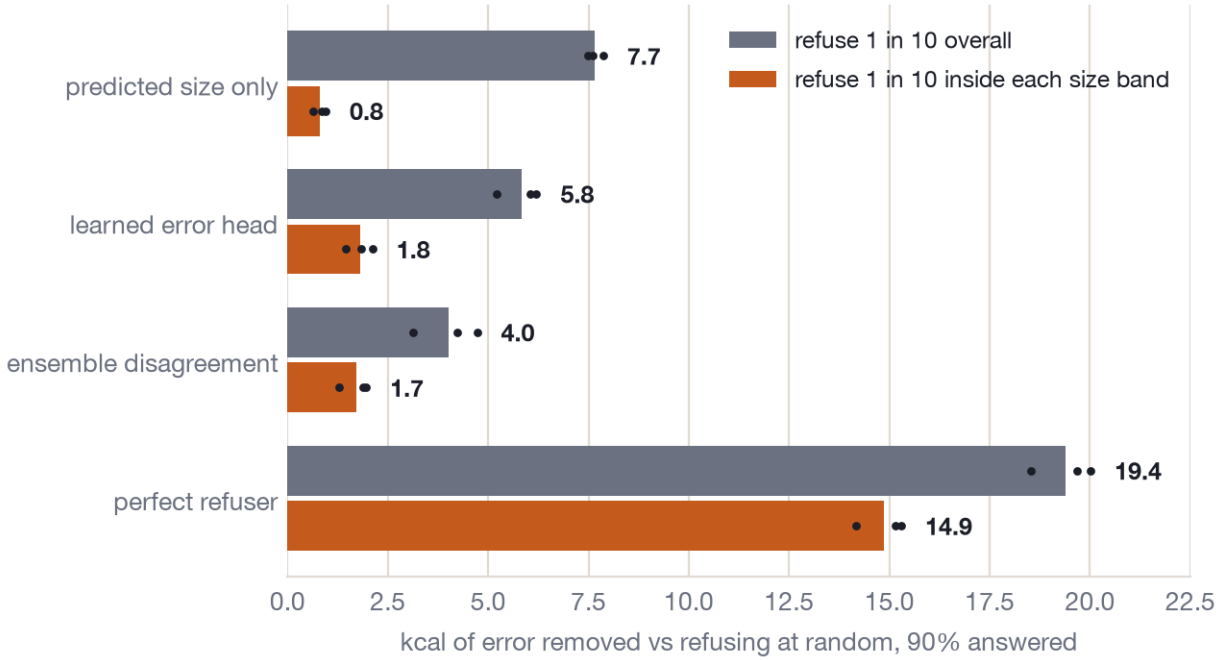
Area under the detection curve: pixel metrics spot damaged photos almost perfectly, yet their rank correlation with the calorie error stays within ±0.08. [ours]

≤ 8%

of a perfect refuser's saving is the most any pixel check could add on all 3,262 clean photos (95% bound; 90% answered, inside size bands). Sharpness points the wrong way: +1.0–1.2 kcal. [ours]

RESULT 2

Most of a gate’s win is refusing big meals.



All 3,262 dishes out of fold; bars: mean of three backbones, dots: each backbone [ours]

Without a size control

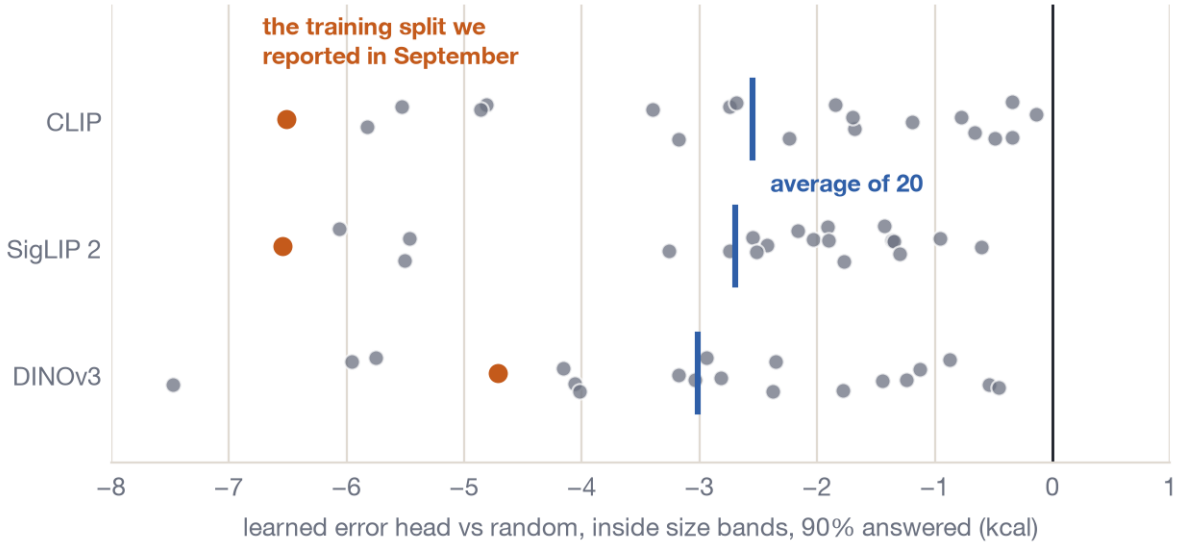
Refusing the meals with the biggest predicted calories removes 7.5–7.9 kcal, more than the learned error head (5.2–6.2). It keeps meals averaging 213–215 kcal; a perfect refuser keeps 224–225.

So every gate is scored twice

The second time, it must refuse the same share inside each of 10 predicted-size bands. Size can no longer be the lever. The size heuristic drops to under 1 kcal; the learned gates keep 1.3–2.1 kcal; a perfect refuser keeps about 15.

CHECKS 1 AND 2: NEW DATA, AND THE MODEL RETRAINED 20 WAYS

Our September headline was a lucky draw.



Error head inside size bands, 90% answered, official test split; each dot is one retraining with different calibration photos [ours]

Retrained 20 ways

The split we reported (–6.5 / –6.5 / –4.7 kcal) was the most favourable of 20 on two backbones and 4th on the third. The average is –2.5 / –2.7 / –3.0.

367 new sessions

Error head –1.3 / –2.6 / –1.9 (significant on two); ensemble disagreement –2.0 / –1.7 / –1.8 kcal (on all three).

Why our intervals missed it

They resampled test photos but held the trained model fixed, so they left out how much the result moves when the model is retrained.

NOT THE EFFECT WE REPORTED the direction holds; the size is about a third of what we said.

CHECKS 3 AND 4: ONE POOLED ESTIMATE, AND PLACEBOS

With size held fixed, a small signal is real.



Pooled

Error head −1.4 / −1.8 / −2.1, ensemble −1.3 / −2.0 / −1.9 kcal: significant on all three; about a tenth of a perfect refuser (14.2–15.3).

Placebo

Heads trained on shuffled errors score about zero (standard deviation ≤ 0.3). The real head sits 4.7–7.3 standard deviations away; none of 200 matched it.

AND IT BEHAVES LIKE A REAL SIGNAL With 20 size bands instead of 10, the size heuristic falls to 0.1–0.7 kcal while the error head holds (−1.5 / −1.8 / −1.9). When the app refuses half the photos, the head removes 6.0–7.9 kcal and the ensemble 4.8–8.9, inside size bands.

All 3,262 dishes, each scored by models that never saw its session (366 clusters); 5 refits averaged; 2,000 session resamples; Holm over 13 gates [ours]

VERDICT

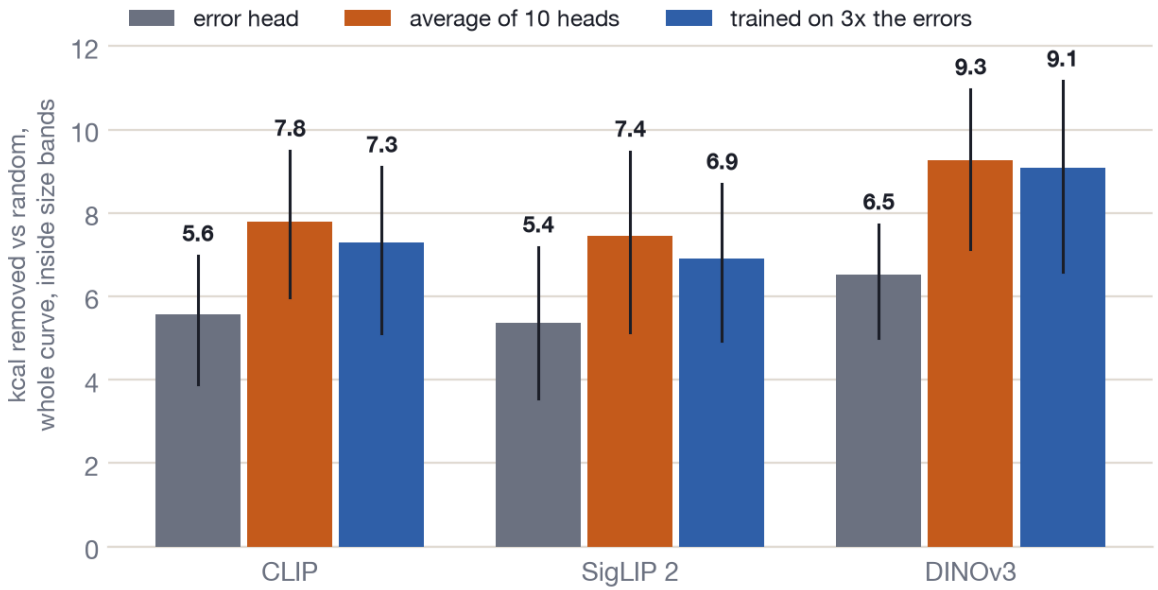
Is it conclusive? Claim by claim.

claim	verdict	evidence
Pixel checks do not tell an app it is guessing.	HOLDS	at most 8% of a perfect refuser (95% bound, three backbones); sharpness points the wrong way
Most of a gate's win is refusing big meals.	HOLDS	size heuristic beats every learned gate without the control
With size held fixed, a difficulty signal remains.	HOLDS, SMALL	about a tenth of a perfect refuser at 90%; 5–7 standard deviations from placebo
"Half the win survives the size control" (September).	OVERSTATED	a favourable training split; the robust value is about a third
The error head beats the size heuristic inside size bands.	PARTLY	over the whole curve on 3 of 3 backbones; at 90% answered on 2 of 3
Which model-aware gate is best.	NOT SEPARABLE	error head and ensemble disagreement tie inside size bands

On new sessions and under retraining, every claim we still make kept its direction. Only the September size shrank.

EXPLORATORY

Two things we tried.



Kcal removed over the whole answer-rate curve, inside size bands; bars: 95% interval [ours]

A steadier error head

The head changed with every retraining, so we averaged 10 heads, each calibrated on different photos (orange). One fitted head becomes steadier (its standard deviation across refits falls 1.00 → 0.67 kcal on CLIP, 0.92 → 0.42 on SigLIP 2) and removes 2.1–2.7 kcal more over the whole curve on all three ($p \leq 0.001$). One head trained on three times the errors (blue) gains 1.5–2.5 kcal. At 90% answered neither changes anything.

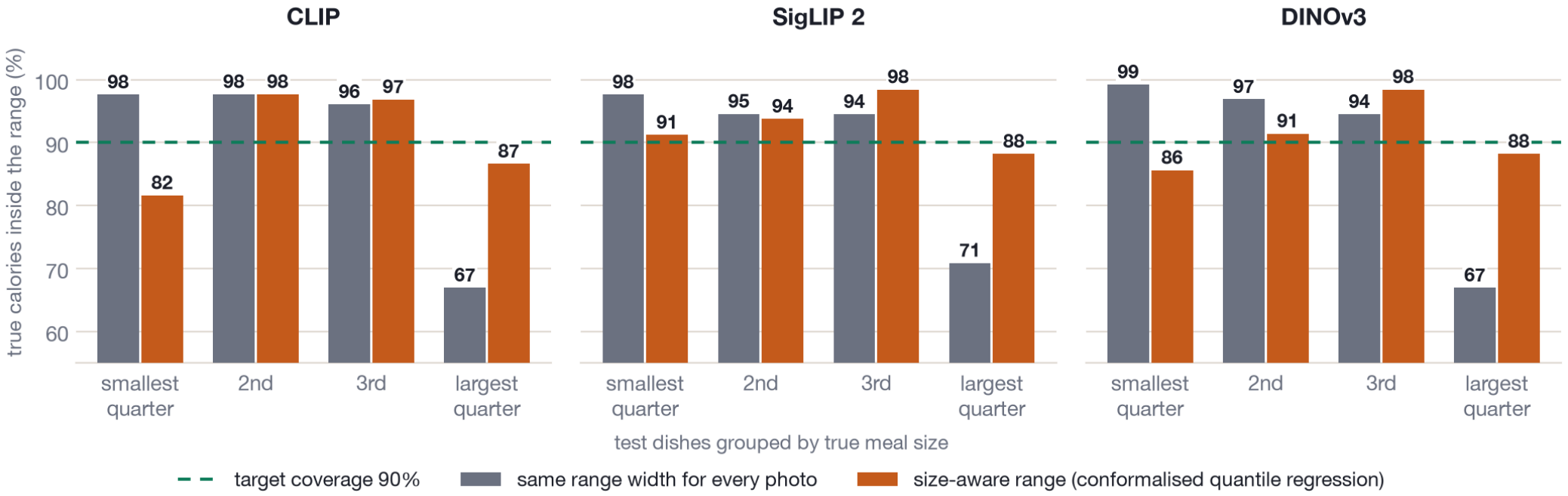
Disagreement across backbones

Best gate for CLIP (–2.6 kcal), weakest model-aware gate for DINOv3 (–0.8): it mostly flags the weaker models' mistakes.

EXPLORATORY, NOT CONFIRMED we designed both after seeing the instability and ran each once, with nothing tuned. They need new photos before we would claim them.

RESULT 3: CONFIDENCE RANGES

Ranges that stay honest for big meals.



Same width for every photo [5]

Covers about 90% overall, but only 67–71% of the largest quarter of meals.

Size-aware ranges, CQR [12]

Widen with the predicted meal: 89–91% of the largest meals over 20 retrainings, 88–90% on new sessions; better interval score. The smallest meals drop to 82–87%.

With the gate

A threshold set on calibration answers 89–91% of test photos; ranges calibrated on answered photos cover 91–94% of them.

What we can and cannot claim.

CONCLUSIONS

1. **Photo quality is not the warning sign.**
   On clean photos no pixel check beats random; at most 8% of a perfect refuser.

2. **Most of a gate's win is meal size.**
   Any reliability benchmark in this domain needs a size control.

3. **A real but small difficulty signal remains.**
   About a tenth of a perfect refuser at 90% answered; more when the app refuses more.

4. **Size-aware ranges keep big meals honest.**
   Small meals pay for it.

LIMITATIONS → FUTURE WORK

One US cafeteria, one overhead camera.
→ Real photos with weighed labels: NutritionVerse-Real, 889 photos [19].

Damage is simulated.
→ Nutrition5k's side cameras: 1,494 more dishes, real viewpoints.

The within-band signal is small.
→ Stronger signals; confirm the steadier head on new photos.

Size-aware ranges under-cover small meals.
→ Ranges that use size and difficulty together.

Labels are sums of weighed ingredients.
→ Ranges on self-reported labels, like real logging: SNAPMe [20].

An app can learn to flag some photos it gets wrong, beyond their size, but only a little; and it can say how wrong it might be, honestly, if the range grows with the meal.

REFLECTION AND CONTRIBUTIONS

What we would do differently.

WORKED WELL

Freezing the predictor: every experiment reruns on a laptop in under two hours.

A control that reads only meal size. It caught a gate that only looked skilful.

Plate sessions as the unit, from September.

WE WOULD DO DIFFERENTLY

Keep a confirmation set aside from day one and vary the training split before presenting a headline. Our September number was about three times too large.

Look for real phone photos earlier.

WHAT WE LEARNED

An interval only covers the randomness you resample.

Before trusting a gate, ask what a perfect refuser would refuse.

Test against placebos: a smooth function of the features finds some signal by luck.

WHO DID WHAT

Nisarg Patel

Did nutrition-estimation papers; survey draft

Presents slides 1–4

Shreya Tigga

Did uncertainty and selective-prediction papers; survey draft

Presents slides 5–7

Keanu De Cleene

Did image-quality and corruption papers; survey draft

Presents slides 8–10

Sushant Krishnan

Did pipeline, experiments and analysis; survey curation

Presents slides 11–13

Kevin Zou

Did conformal-prediction papers; survey draft

Presents slides 14–16

CLOSING

Sources, AI use, and the code.

[1] Q. Thames, A. Karpur, W. Norris, F. Xia, L. Panait, T. Weyand, and J. Sim, “Nutrition5k: Towards automatic nutritional understanding of generic food,” in Proc. IEEE/CVF CVPR, 2021, pp. 8899–8907.
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All figures and tables are ours unless tagged; photos on slide 2 are Nutrition5k dishes [1], reproduced under the dataset licence.

GENERATIVE AI USE Claude Code (Anthropic) was used throughout, under the group’s direction: it wrote most of the pipeline and analysis code, proposed and ran the robustness checks after we asked how to answer the update feedback, and drafted the figures, these slides and the speaker notes. The group chose the direction and which checks to run, and is responsible for every claim; every number on these slides is read from the results files by the deck builder.

CODE AND RESULTS **github.com/sushantkrishnan/seeing-what-you-eat** · bash reproduce.sh reruns everything; code/README.md says what is ours